

3.1 FACTORS AND MULTIPLES

Define a factor as a number which divides the dividend completely leaving no remainder.

We know that if a dividend is divided by a divisor, we get quotient and remainder.

Let us consider division of 8 by 2 and 3 separately.

Division by 2

$$\begin{array}{r} 2 \overline{) 8} \quad 4 \\ - 8 \\ \hline 0 \end{array} \quad \text{or} \quad 8 \div 2 = 4$$

remainder = 0

Division by 3

$$\begin{array}{r} 3 \overline{) 8} \quad 2 \\ - 6 \\ \hline 2 \end{array} \quad \text{or} \quad 8 \div 3 = 2$$

remainder = 2

Here 2 is a factor of 8 because remainder is zero or 2 divides 8 exactly. 3 is not a factor of 8 because remainder is not zero or 3 does not divide 8 exactly.

A factor is a number which divides the dividend completely leaving no remainder.

Example: Which of the following are the factors of 10.

(a) 2

(b) 4

(c) 5

(d) 6

Solutions:

$$\begin{array}{r} (a) \quad 2 \overline{) 10} \quad 5 \\ - 10 \\ \hline 0 \end{array} \quad 10 = 2 \times 5$$

Since Remainder = 0
So, 2 is a factor of 10

$$\begin{array}{r} (b) \quad 4 \overline{) 10} \quad 2 \\ - 8 \\ \hline 2 \end{array}$$

Since Remainder is not zero
So, 4 is not a factor of 10

$$\begin{array}{r} (c) \quad 5 \overline{) 10} \quad 2 \\ - 10 \\ \hline 0 \end{array} \quad 10 = 5 \times 2$$

Since Remainder = 0
So, 5 is a factor of 10

$$\begin{array}{r} (d) \quad 6 \overline{) 10} \quad 1 \\ - 6 \\ \hline 4 \end{array}$$

Since Remainder is not zero
So, 6 is not a factor of 10

So, 2 and 5 are the factors of 10.

Example 2: Find all the factors of:

(a) 12

(b) 20

Solution:

(a) 12

$$\text{As } 12 = 1 \times 12$$

$$12 = 2 \times 6$$

$$12 = 3 \times 4$$

So, 1, 2, 3, 4, 6 and 12 can divide 12 exactly.

Hence 1, 2, 3, 4, 6 and 12 are factors of 12.

(b) 20

$$\text{As } 20 = 1 \times 20$$

$$20 = 2 \times 10$$

$$20 = 4 \times 5$$

So, 1, 2, 4, 5, 10 and 20 can divide 20 exactly.

Hence 1, 2, 4, 5, 10 and 20 are factors of 20.

The number itself and 1 are always the factors of the given number.

Define a multiple as a dividend into which a factor can divide

Let us consider the division of 18 by 2.

$$\begin{array}{r} 2 \overline{) 18} \quad 9 \\ - 18 \\ \hline 0 \end{array}$$

$$18 = 2 \times 9$$

Here the divisor 2 is called factor of 18 because remainder is zero.

The dividend 18 is called multiple of 2 because 2 is factor of 18.

Thus

Multiple of a given number is the dividend, for which the given number is its factor.

So, a number has always unlimited number of multiples.

Example 1: Is 20 a multiple of 3?

Solution:

$$\begin{array}{r} 3 \overline{) 20} \text{ (6)} \\ - 18 \\ \hline 2 \end{array}$$

Since 3 is not a factor of 20
So, 20 is not a multiple of 3.

Example 2: Find all multiples of 2.

Solution: Multiples of 2 can be found as:

$$\begin{aligned} 2 \times 1 &= 2 \\ 2 \times 2 &= 4 \\ 2 \times 3 &= 6 \\ 2 \times 4 &= 8 \\ &\text{and so on} \end{aligned}$$

So, the multiples of 2 are 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, ...

Example 3: Find the multiples of 3 less than 15.

Solution: The multiples of 3 are 3, 6, 9, 12, 15, 18, ...

So, the multiples less than 15 are 3, 6, 9, 12

Example 4: Find the multiples of 4 which are between 9 and 23.

Solution: The multiples of 4 are 4, 8, 12, 16, 20, 24, 28, ...

So, the multiples between 9 and 23 are 12, 16, 20.

Define even numbers as the numbers which are multiples of 2

Any number which is multiple of 2, is called an even number.

As 2, 4, 6, 8, 10, 12, ... are multiples of 2.

So, 2, 4, 6, 8, 10, 12, ... are even numbers.

In other words, an even number is a number which is exactly divisible by 2.

Rule: A number having 0, 2, 4, 6, or 8 at its ones place is an even number.

Example: Which of the following numbers are even:

- (a) 257 (b) 7208 (c) 11114 (d) 15683

Solution:

- (a) 257 is not even because digit at ones place is 7.
 (b) 7208 is even because digit at ones place is 8.
 (c) 11114 is even because digit at ones place is 4.
 (d) 15683 is not even because digit at ones place is 3.

Define odd numbers as the numbers which are not multiples of 2

A number which is not multiple of 2 is called an odd number.
 As 1, 3, 5, 7, 9, 11, 13, ... are not multiples of 2.

So, 1, 3, 5, 7, 9, 11, 13, 15, ... are odd numbers.

In other words

An odd number is a number which is not exactly divisible by 2.

Rule: A number having 1, 3, 5, 7 or 9 at its ones place is an odd number.

Example: Identify the odd or even numbers as the multiple of 2 and explain.

- (a) 3145 (b) 6784 (c) 9210 (d) 2461

Solution:

	Number	Type	Explanation
(a)	3145	odd	As digit at ones place is 5, which is not even number. Hence, 3145 is odd number.
(b)	6784	even	As digit at ones place is 4, which is even number. So, 6784 is an even number.
(c)	9210	even	As digit at ones place is 0, which is even number. So, 9210 is even number.
(d)	2461	odd	As digit at ones place is 1, which is not even number. So, 2461 is odd number.

Define prime numbers as numbers which have only two factors (i.e., 1 and itself)

A natural number which has only two distinct factors 1 and number itself is called a prime number.

As 2, 3, 5, 7, 11, 13, 17, 19, ... have only two factors.

So, they are prime numbers.

Example: Which of the following numbers is a prime number?

- (a) 23 (b) 8

Solution:

- (a) 23 has two factors 1 and 23. So, 23 is prime number.
(b) 8 has four factors 1, 2, 4, 8. So, 8 is not a prime number.

Define composite numbers as numbers which have more than two factors

A natural number which has more than two factors is called a composite number.

As 4, 6, 8, 9, 10, 12, 14, 15, ... have more than two factors.

So, they are composite numbers.

Example: Identify the composite or prime numbers with reasons.

- (a) 8 (b) 17 (c) 9 (d) 5

Solution:

	Number	Type	Reasons
(a)	8	Composite number	As 1, 2, 4, 8 are the factors of 8, which are more than two. Hence it is composite number.
(b)	17	Prime number	As 1, 17 are the only two distinct factors so, 17 is prime number.
(c)	9	Composite number	As 1, 3, 9 are the factors of 9, which are more than two. Hence it is composite number.
(d)	5	Prime number	As 1, 5 are the only two distinct factors of 5 so, 5 is prime number.

Know that 1 is neither prime nor composite as it has only one factor which is 1 itself

According to the number of factors, there are three types of natural numbers as given below:

Natural Number	Type	Factors
1	Neither prime nor composite	1
2	Prime number	1, 2
3	Prime number	1, 3
4	Composite number	1, 2, 4
5	Prime number	1, 5
6	Composite number	1, 2, 3, 6

So 2, 3, 5, 7, 11, 13, ... are prime numbers.

4, 6, 8, 9, 10, 12, 14, 15, ... are composite numbers and 1 is neither prime nor composite number because it has only one factor which is 1 itself.

Example: Identify prime, composite and the number which is neither prime nor composite from the following:

5, 8, 16, 1, 19

Solution:

Prime numbers are 5 and 19

Composite numbers are 8 and 16

The number which is neither prime nor composite is 1

Know that 1 is a factor of every number.

If we divide any number by 1, the remainder will always be zero. as it is clear from the following examples:

- (1) 1 is factor of 5
- (2) 1 is factor of 8
- (3) 1 is factor of 104

So, 1 is factor of every number because it produces 0 as remainder.

Example: Find factors of 2, 5 and 6 and which number is factor of each of 2, 5 and 6.

Solution:

Factors of 2 are 1, 2

Factors of 5 are 1, 5

Factors of 6 are 1, 2, 3, 6

Here '1' is the only number which is factor of each of 2, 5 and 6.

Know that 2 is the only even prime number whereas all other prime numbers are odd

We know that even natural numbers are 2, 4, 6, 8, 10, ...

From these numbers 2 has only two factors which are 1 and 2.

Whereas all other even numbers have more than two factors.

Hence 2 is the only even number which is also prime.

We know that all prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, ...
From these prime numbers 2 is only even prime number and all other are odd.

EXERCISE 3.1

1. Which of the following are the factors of 20.

- (a) 2 (b) 8 (c) 5 (d) 3

2. Write all the factors of each of the following.

- (a) 15 (b) 30 (c) 50 (d) 125 (e) 150

3. Which of the following are multiples of 6.

- (a) 10 (b) 18 (c) 96 (d) 200

4. Write first seven multiples of:

- (a) 4 (b) 7 (c) 12 (d) 15 (e) 20

5. Write prime numbers between:

- (a) 6 and 15 (b) 20 and 40 (c) 60 and 95

6. Write an even prime number.

7. Which of the following is even or odd?

- (a) 10 (b) 47 (c) 75 (d) 91 (e) 100 (f) 117

8. Write all the even numbers between 21 and 51.

9. Write all the odd numbers between 10 and 40.

10. Write all the composite numbers less than 40.

11. Write all the composite numbers between 71 and 101.

12. Write all the prime numbers between 20 and 60.

13. Why 1 is neither prime nor composite?

3.2 TESTS FOR DIVISIBILITY

Test by inspection whether the numbers 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 15 and 25 can divide a given number.

There are rules by which we can check whether a given number is exactly divisible by 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 15 and 25. These rules are called tests for divisibility.

Test of Divisibility by 2 :

A number is exactly divisible by 2 if the digit at units place of the number is 0, 2, 4, 6 or 8.

Example: 50, 62, 724, 9126, 3338 are exactly divisible by 2.

Test of Divisibility by 3 :

A number is exactly divisible by 3 if the sum of the digits of the number is divisible by 3.

Example: 24510 is exactly divisible by 3 because $2 + 4 + 5 + 1 + 0 = 12$ and 12 is divisible by 3

Test of Divisibility by 4 :

A number is divisible by 4 if the number formed by the last two digits of the number is divisible by 4 or the two digits are zeros.

Example: 1002316 is divisible by 4 because 16 is divisible by 4.

Test of Divisibility by 5 :

A number is divisible by 5 if digit at the units place of the number is 0 or 5.

Example: 5210 and 4115 are exactly divisible by 5.

Test of Divisibility by 6 :

A number is divisible by 6 if the number is exactly divisible by both 2 and 3.

Example: 2142 is divisible by 6 because it is exactly divisible by both 2 and 3.

Test of Divisibility by 8 :

A number is exactly divisible by 8 if the number formed by the last three digits is exactly divisible by 8 or the last three digits are zero.

Example: 213832 is divisible by 8 because 832 is exactly divisible by 8.

Test of Divisibility by 9 :

A number is exactly divisible by 9 if the sum of the digits of the number is exactly divisible by 9.

Example:

9241011 is exactly divisible by 9 because $9 + 2 + 4 + 1 + 0 + 1 + 1 = 18$ and 18 is exactly divisible by 9.

Test of Divisibility by 10 :

A number is exactly divisible by 10 if the digit at units place of the number is zero.

Example: 25670 is exactly divisible by 10.

Test of Divisibility by 11 :

A number is exactly divisible by 11 if the difference of the sum of digits at odd places from the sum of its digits at even places is either 0 or exactly divisible by 11.

Example:

7546 is exactly divisible by 11 because $(7 + 4) - (5 + 6) = 11 - 11 = 0$
and 907665 is divisible by 11 because $(9 + 7 + 6) - (0 + 6 + 5) = 22 - 11 = 11$ which is exactly divisible by 11.

Test of Divisibility by 12 :

A number is exactly divisible by 12 if the number is exactly divisible by both 3 and 4.

Example: 234084 is exactly divisible by 12 because it is exactly divisible by 3 and 4 (As $2 + 3 + 4 + 0 + 8 + 4 = 21$ is divisible by 3 and 84 is exactly divisible by 4).

Test of Divisibility by 15 :

A number is exactly divisible by 15 if the number is exactly divisible by both 3 and 5

Example: 26130 is divisible by 15 because it is divisible by both 3 and 5.

Test of Divisibility by 25 :

A number is exactly divisible by 25 if the number formed by last two digits is exactly divisible by 25 or the last two digits are zeros.

Example: 2341625 is divisible by 25 because 25 is divisible by 25.

Example: Test whether:

- (i) 2412006 is exactly divisible by 3
- (ii) 6123816 is exactly divisible by 8
- (iii) 61710 is exactly divisible by 6
- (iv) 36578146 is exactly divisible by 11
- (v) 43210284 is exactly divisible by 12
- (vi) 32412075 is exactly divisible by 15

Solution:

- (i) $2 + 4 + 1 + 2 + 0 + 0 + 6 = 15$. Since 15 is divisible by 3.
So, 2412006 is divisible by 3.
- (ii) 6123816 is divisible by 8. Since 816 is divisible by 8.
- (iii) $6 + 1 + 7 + 1 + 0 = 15$. Since 15 is divisible by 3 and digit at units place is 0, so it is divisible by 2.
Hence 61710 is divisible by 6.
- (iv) Here $(3 + 5 + 8 + 4) - (6 + 7 + 1 + 6) = 20 - 20 = 0$
So, 36578146 is exactly divisible by 11
- (v) As $4 + 3 + 2 + 1 + 0 + 2 + 8 + 4 = 24$ is exactly divisible by 3 and 24 is also exactly divisible by 4
So, 43210284 is exactly divisible by 3 and 4
Hence it is exactly divisible by 12
- (vi) As $3 + 2 + 4 + 1 + 2 + 0 + 7 + 5 = 24$ is exactly divisible by 3 and there is 5 at ones place
So, 32412075 is exactly divisible by 3 and 5
Hence it is exactly divisible by 15

EXERCISE 3.2**1. Test whether.**

- (i) 251061 is exactly divisible by 3
- (ii) 92348 is exactly divisible by 4
- (iii) 49230 is exactly divisible by 9
- (iv) 24150 is exactly divisible by 8
- (v) 10005 is exactly divisible by 2
- (vi) 241566 is exactly divisible by 3
- (vii) 34268 is exactly divisible by 2
- (viii) 241361 is exactly divisible by 4
- (ix) 5123864 is exactly divisible by 8
- (x) 4158720 is exactly divisible by 9
- (xi) 210006 is exactly divisible by 2
- (xii) 1234562 is exactly divisible by 3

2. Which of the following numbers are exactly divisible by 6?

- (i) 2456 (ii) 7121700 (iii) 19206

3. Which of the following numbers are exactly divisible by 5?

- (i) 24567 (ii) 230590 (iii) 111165

4. Which of the following numbers are exactly divisible by 10?

- (i) 25670 (ii) 123600 (iii) 14675

5. Which of the following numbers are exactly divisible by 11?

- (i) 215762 (ii) 52958400 (iii) 94857290

6. Which of the following numbers are exactly divisible by 12?

- (i) 245168 (ii) 512100 (iii) 2561742

7. Which of the following numbers are exactly divisible by 15?
(i) 51300 (ii) 523449 (iii) 236790
8. Which of the following numbers are exactly divisible by 25?
(i) 2563400 (ii) 341236 (iii) 4102375
9. Which of the following numbers are exactly divisible by 2, 3, and 6?
(i) 14562 (ii) 1101361 (iii) 518214
10. Which of the following numbers are exactly divisible by 4 and 8?
(i) 513864 (ii) 617116 (iii) 2415848
11. Which of the following numbers are exactly divisible by 2, 5 and 10?
(i) 120340 (ii) 210785 (iii) 412360

3.3 FACTORIZATION

The process of factorizing a number into its factors is called factorization.

Rule: Product of factors of a given number is always equal to the number.

For example: (i) $8 = 2 \times 4$ (ii) $12 = 6 \times 2$
Here 2 and 4 are Here 6 and 2 are
factors of 8 factors of 12

Define prime factorization as the process of factorizing a number into its prime factors

The process of factorizing a number into its prime factors is called prime factorization.

For example: (i) $8 = 2 \times 2 \times 2$ (ii) $12 = 2 \times 2 \times 3$

Prime factorization can be done by two methods as explained in the following example.

Example 1: Find the prime factorization of 60:

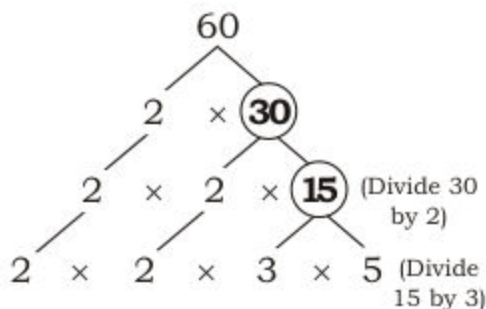
Solution:

Division method

2	60
2	30
3	15
5	5
	1

So, $60 = 2 \times 2 \times 3 \times 5$
which is prime factorization
of 60

Tree method



So, $60 = 2 \times 2 \times 3 \times 5$
which is prime factorization
of 60

Example 2: Find the prime factorization of 540:

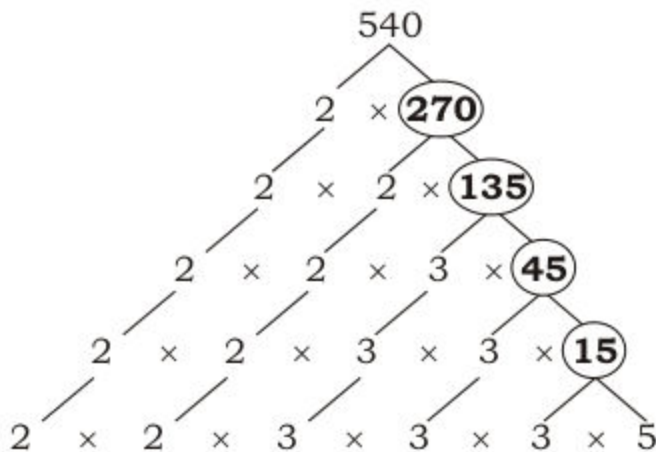
Solution:

Division method

2	540
2	270
3	135
3	45
3	15
5	5
	1

So,
 $540 = 2 \times 2 \times 3 \times 3 \times 3 \times 5$
which is prime factorization
of 540.

Tree method



So,
 $540 = 2 \times 2 \times 3 \times 3 \times 3 \times 5$
which is prime factorization of 540.

Teacher's Note

Teacher should help the students to practice both methods of prime factorization.

Recognize index notation

Consider the factors of 360.

2	360
2	180
2	90
3	45
3	15
5	5
	1

$$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$$

Here, factor 2 is 3 times $= 2^3$ (index = 3)

factor 3 is 2 times $= 3^2$ (index = 2)

and factor 5 is 1 time $= 5^1$ (index = 1)

So, prime factorization of 360 can be expressed as

$$360 = 2^3 \times 3^2 \times 5^1$$

The prime factorization of 360 in the above form is known as index notation.

Factorize a given number and express its factors in the index notation

Let us factorize a given number and express its factors in the index notation with the help of an example.

Example: Find prime factorization and express it in index notation.

(i) 432

(ii) 1500

Solution:

(i) 432

2	432
2	216
2	108
2	54
3	27
3	9
3	3
	1

Hence,

$$\begin{aligned} 432 &= 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \\ &= 2^4 \times 3^3 \end{aligned}$$

(ii) 1500

2	1500
2	750
3	375
5	125
5	25
5	5
	1

Hence,

$$\begin{aligned} 1500 &= 2 \times 2 \times 3 \times 5 \times 5 \times 5 \\ &= 2^2 \times 3^1 \times 5^3 \end{aligned}$$

EXERCISE 3.3

1. **Write in index notation.**

(i) $2 \times 2 \times 2 \times 3 \times 5 \times 3$

(ii) $3 \times 5 \times 5 \times 5 \times 7 \times 7$

(iii) $2 \times 3 \times 3 \times 7 \times 7 \times 3 \times 3$

2. **Find the prime factorization by division method and express in index notation.**

(i) 24

(ii) 48

(iii) 216

(iv) 250

(v) 468

(vi) 540

(vii) 1024

(viii) 5000

3. **Can you write factorization of a prime number in index notation.**

3.4 HCF (Highest Common Factor)

Define HCF as the greatest number which is a common factor of two or more numbers.

HCF of two or more given numbers is the common factor of the given numbers which is the greatest of all the common factors.

Example: Find HCF of 24 and 40.

Solution:

Factors of 24 are: 1, 2, 3, 4, 6, 8, 12, 24

Factors of 40 are: 1, 2, 4, 8, 10, 20, 40

Common factors are: 1, 2, 4, 8

The greatest common factor is 8

Hence, $\text{HCF} = 8$

Find HCF of two or more than two numbers by prime factorization and long division method

There are two main methods to find HCF. These methods are explained below:

(i) Prime Factorization Method:

In this method, we use the following rule:

HCF = Product of common factors of two or more numbers.

Example: Find HCF of 48, 36 and 40.

Solution:

2	48
2	24
2	12
2	6
3	3
	1

2	36
2	18
3	9
3	3
	1

2	40
2	20
2	10
5	5
	1

Here

$$\begin{aligned}
 48 &= 2 \times 2 \times 2 \times 2 \times 3 \\
 36 &= 2 \times 2 \times 3 \times 3 \\
 40 &= 2 \times 2 \times 2 \times 5
 \end{aligned}$$

Common factors are 2 and 2

$$\begin{aligned}
 \text{So, HCF} &= 2 \times 2 \\
 &= 4
 \end{aligned}$$

(ii) Long Division Method:

In case of two numbers, first find the HCF by the method as given below:

- Divide greater number by the smaller number.
- Take remainder as divisor and divide the first divisor by it.
- Again remainder will be taken as divisor and 2nd divisor will be treated as dividend.
- Continue this process till remainder becomes zero.
- The last divisor will be HCF.

Note: In case of three numbers, we first find HCF of any two numbers and then continue the process on that HCF and the remaining number till that last divisor, which is HCF of the all three numbers.

Example 1:

Find HCF of 24 and 64 by long division method.

Solution:

$$\begin{array}{r}
 24 \overline{) 64} \begin{array}{l} 2 \\ - 48 \\ \hline 16 \end{array} \\
 16 \overline{) 24} \begin{array}{l} 1 \\ - 16 \\ \hline 8 \end{array} \\
 \text{HCF} \longrightarrow 8 \overline{) 16} \begin{array}{l} 2 \\ - 16 \\ \hline 0 \end{array}
 \end{array}$$

So, HCF = 8

Example 2:

Find HCF of 16, 36 and 70.

Solution: First we find HCF of 16 and 36.

$$\begin{array}{r}
 16 \overline{) 36} \begin{array}{l} 2 \\ - 32 \\ \hline 4 \end{array} \\
 4 \overline{) 16} \begin{array}{l} 4 \\ - 16 \\ \hline 0 \end{array}
 \end{array}$$

So, HCF of 16 and 36 = 4

Now we find HCF of 4 and 70.

$$\begin{array}{r}
 4 \overline{) 70} \begin{array}{l} 17 \\ - 4 \\ \hline 30 \\ - 28 \\ \hline 2 \end{array} \\
 \text{HCF} \longrightarrow 2 \overline{) 4} \begin{array}{l} 2 \\ - 4 \\ \hline 0 \end{array}
 \end{array}$$

Hence, the HCF of 16, 36 and 70 is 2.

EXERCISE 3.4

- Find the HCF of the following numbers by prime factorization method.
(i) 50, 75 (ii) 98, 196 (iii) 144, 198
(iv) 120, 144, 204 (v) 106, 159, 265
(vi) 12, 48, 36, 24 (vii) 60, 70, 420, 480
- Find HCF of the following numbers by long division method.
(i) 12, 20 (ii) 81, 117 (iii) 935, 1320
(iv) 252, 576 (v) 2241, 8217, 747
(vi) 30, 120, 90, 210
- What is the HCF of two prime numbers?
(Take any two prime numbers)

3.5 LCM (Least Common Multiple)

Define LCM as the smallest number which is a common multiple of two or more numbers.

LCM of two or more given numbers is the smallest of all their common multiples.

Example: Find the LCM of 3 and 4.

Solution:

Multiples of 3 are: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, ...

Multiples of 4 are: 4, 8, 12, 16, 20, 24, 28, 32, 36, ...

Common multiples of 3 and 4 are: 12, 24, 36, ...

The smallest common multiple = 12

i.e. LCM = 12

Find LCM of two or more number by prime factorization and division method

There are two main methods to find LCM. These are explained below:

(i) Prime Factorization Method:

In this method first we find prime factors of each given number then we find LCM by the following rule.

i.e. $\text{LCM} = \text{Product of common and non-common factors.}$

Note: (i) In case of three numbers, we consider a factor as common factor if it is common factor of two numbers or of three numbers.

(ii) Same rule will be applied for more than three numbers.

Example: Find LCM of 12, 18, 24.

Solution:

$$\begin{array}{r|l} 2 & 12 \\ \hline 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 18 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 24 \\ \hline 2 & 12 \\ \hline 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

Here

$$\begin{array}{lcl} 12 & = & 2 \times 2 \times 3 \\ 18 & = & 2 \times 3 \times 3 \\ 24 & = & 2 \times 2 \times 2 \times 3 \end{array}$$

Here

Common factors are: 2, 2, 3

Non-common factors are: 2, 3

$$\begin{aligned} \text{So, LCM} &= 2 \times 2 \times 3 \times 2 \times 3 \\ &= 72 \end{aligned}$$

(ii) Division Method:

The main points of this method are given as under.

1. Choose the smallest divisor that can divide at least one of given numbers.
2. Write quotient and the undivided numbers in the second row.
3. Again choose the smallest divisor that can divide at least one of them
4. Continue this process until you get "1" as quotient for each number i.e. (Last row contains all ones)

Then $\text{LCM} = \text{Product of all divisors}$.

Example 1:

Find LCM of 3, 4, 5.

Solution:

2	3, 4, 5
2	3, 2, 5
3	3, 1, 5
5	1, 1, 5
	1, 1, 1

$$\begin{aligned}\text{Here LCM} &= 2 \times 2 \times 3 \times 5 \\ &= 60\end{aligned}$$

Example 2:

Find LCM of 12, 15, 50.

Solution:

2	12, 15, 50
2	6, 15, 25
3	3, 15, 25
5	1, 5, 25
5	1, 1, 5
	1, 1, 1

$$\begin{aligned}\text{Here LCM} &= 2 \times 2 \times 3 \times 5 \times 5 \\ &= 300\end{aligned}$$

Relationship between HCF and LCM

Consider the two number **16** and **24**.

Their product $16 \times 24 = 384$

$\text{HCF} = 8$ and $\text{LCM} = 48$

Now multiply HCF and LCM

$$8 \times 48 = 384$$

Thus

The product of any two non zero numbers = The product of HCF and LCM

Example: The product of two numbers is 300 and their LCM is 60. What is their HCF?

Solution: The product of two numbers = 300

$$\text{LCM} = 60, \text{HCF} = ?$$

We know that

$$\text{LCM} \times \text{HCF} = \text{Product of two numbers}$$

$$\text{So, } \text{LCM} \times \text{HCF} = 300$$

$$60 \times \text{HCF} = 300$$

$$\text{HCF} = \frac{300}{60}$$

$$= 5$$

Thus $\text{HCF} = 5$

EXERCISE 3.5

- Using prime factorization, find the LCM of the following numbers.**

(i) 12, 25 and 40	(ii) 21, 49 and 63
(iii) 144, 180 and 384	(iv) 108, 135 and 162
(v) 35, 65 and 75	(vi) 24, 36, 48 and 72
- Calculate the LCM of the following numbers by division method.**

(i) 45 and 55	(ii) 21, 35 and 70
(iii) 24, 40 and 60	(iv) 72, 108 and 120
- The product of two numbers is 360. If their HCF is 6, what is their LCM?**

3.6 Applications of HCF and LCM

Solve real life problems related to HCF and LCM

HCF and LCM are used in daily life as explained in the following examples.

Example 1:

Find the greatest length of a measuring tape which can be used to measure exactly 520 cm and 360 cm.

Solution: The greatest length of the tape is the HCF of 520 cm and 360 cm.

Lets find the required HCF by division method.

$$\begin{array}{r}
 360 \overline{) 520} \begin{array}{l} 1 \\ - 360 \\ \hline 160 \end{array} \\
 160 \overline{) 360} \begin{array}{l} 2 \\ - 320 \\ \hline 40 \end{array} \\
 40 \overline{) 160} \begin{array}{l} 4 \\ - 160 \\ \hline 0 \end{array} \\
 \times
 \end{array}$$

The HCF of 520 and 360 is 40

Hence, the required length of the tape is 40 cm.

Example 2:

Find the greatest length of wooden scale which can be used to measure 63 cm, 85 cm and 47 cm leaving remainders 3 cm, 5 cm and 7 cm respectively.

Solution:

$$\begin{array}{r}
 63, 85, 47 \\
 -3, -5, -7 \\
 \hline
 60, 80, 40
 \end{array}$$

Here, we have to find HCF
of 60, 80 and 40

Lets find the HCF by division method.

$$\begin{array}{r} 60 \overline{) 80} (1 \\ \underline{- 60} \\ 20 \end{array}$$

$$\begin{array}{r} 20 \overline{) 60} (3 \\ \underline{- 60} \\ 0 \end{array}$$

As HCF of 60 and 80 is 20

Now we find HCF of 20 and 40

$$\begin{array}{r} 20 \overline{) 40} (2 \\ \underline{- 40} \\ 0 \end{array}$$

As the HCF of 60, 80, 40 is 20

So, the required length of wooden scale is 20 cm.

Example 3: Find the least quantity of sugar which can be exactly measured by 100 gm, 150 gm or 200 gm weights.

Solution: The required least quantity of sugar can be found with the help of LCM.

$$\begin{aligned} \text{Here LCM} &= 2 \times 2 \times 2 \times 3 \times 5 \times 5 \\ &= 600 \end{aligned}$$

Hence the required quantity of sugar is 600 gm.

2	100, 150, 200
2	50, 75, 100
2	25, 75, 50
3	25, 75, 25
5	25, 25, 25
5	5, 5, 5
	1, 1, 1

Example 4: Find the least number of children which can stand in rows of 10, 30 and 60 children such that each row has 7 children short.

Solution: First of all we find LCM of 10, 30 and 60.

2	10, 30, 60
2	5, 15, 30
3	5, 15, 15
5	5, 5, 5
	1, 1, 1

$$\begin{aligned} \text{LCM} &= 2 \times 2 \times 3 \times 5 \\ &= 60 \end{aligned}$$

So, the required least number of children = $60 + 7 = 67$

Example 5: Three bells ring at intervals of 10, 30 and 40 minutes respectively. At what time will they ring together if they start ringing simultaneously at 9 a.m.

Solution: First of all we find LCM of 10, 30 and 40.

2	10, 30, 40
2	5, 15, 20
2	5, 15, 10
3	5, 15, 5
5	5, 5, 5
	1, 1, 1

$$\begin{aligned}\text{Here LCM} &= 2 \times 2 \times 2 \times 3 \times 5 \\ &= 120\end{aligned}$$

They will ring together after 120 minutes together i.e. 120 minutes = 2 hours

$$\begin{aligned}\text{Hence the required time} &= (9 + 2) \text{ a.m.} \\ &= 11 \text{ a.m.}\end{aligned}$$

EXERCISE 3.6

- Find the greatest length of a wooden scale which can be used to measure 540 cm and 360 cm exactly.
- Find the greatest number which divides 232 and 305 leaving 7 and 5, respectively as remainder.
- Find the largest number which divides 245 and 1029 leaving remainder 5 in each case.
- Two tankers contain 600 litres and 570 litres of petrol respectively. Find the maximum capacity of the container which can measure the petrol of either tanker in exact number of times.
- The length, breadth and height of a room are 8 m, 6 m and 4 m respectively. Determine the longest tape which can measure the three dimensions of the room exactly.
- Three bells ring at an interval of 5, 10 and 15 minutes respectively. If these started ringing at 8 A.M. Find the time when they will again ring together.

7. Find the least number which when divided by 30, 45 and 60 leaves 9 as remainder.
8. Find the least quantity of milk which can be exactly measured by buckets of capacity 12 litres, 16 litres and 24 litres.
9. The traffic lights at three different road-crossings change after every 48 seconds, 72 seconds and 108 seconds respectively. If they change simultaneously at 7 a.m, after what time will they change again simultaneously?

REVIEW EXERCISE 3

1. Write all the factors of (i) 60 (ii) 250
2. Write first five multiples of (i) 13 (ii) 20
3. Write all prime numbers between 1 and 50.
4. Write all composite numbers between 30 and 60.
5. Which of the following are divisible by 2:
(i) 31621 (ii) 7008 (iii) 91130 (iv) 5178
6. Which of the following are divisible by 3:
(i) 51237 (ii) 30001 (iii) 1001001 (iv) 56712
7. Which of the following are divisible by 4:
(i) 2173 (ii) 41524 (iii) 71611 (iv) 40048
8. Which of the following are divisible by 5:
(i) 2010 (ii) 31625 (iii) 7128 (iv) 1001
9. Which of the following are divisible by 10:
(i) 2165 (ii) 71230 (iii) 10000 (iv) 25618
10. Find prime factors of the following by division method and tree method. Also write in index notation.
(i) 450 (ii) 720

- 11.** Find HCF and LCM by factorization method:
(i) 18, 24 (ii) 12, 15, 40
- 12.** Find HCF and LCM by division method:
(i) 65 and 80 (ii) 26, 65, 169
- 13.** Find the greatest number, that will divide 42, 51 and 67 leaving remainders 2, 1 and 7 respectively.
- 14.** Find the least number which when divided by 20, 30 and 45 leaving 6 as remainder in each case.

SUMMARY

- A factor is a number which divides the dividend exactly.
- A number has limited number of factors.
- Multiple of a given number is the dividend for which the given number is its factor.
- A number has unlimited number of multiples.
- A number which is multiple of 2 is called even number.
- A number which is not multiple of 2 is called an odd number.
- A natural number which has only two distinct factors is called prime number.
- A natural number which has more than two factors is called composite number.
- 1 is neither prime nor composite.
- 1 is factor of every number.
- 2 is the only even number which is also a prime number.

- Divisibility Test a number is divisible by
 - (i) 2 if digit at unit place is 0, 2, 4, 6 or 8.
 - (ii) 3 if the sum of digit is multiple of 3.
 - (iii) 4 if the number formed by last two digit is divisible by 4.
 - (iv) 5 if the digit at unit place is 0 or 5.
 - (v) 6 if the number is divisible by both 2 and 3.
 - (vi) 8 if the number formed by last three digits is divisible by 8.
 - (vii) 9 if sum of digits is divisible by 9.
 - (viii) 10 if digit at units place is zero.
 - (ix) 11 if the difference of sum of digits at odd places from the sum of digits at even places is either 0 or multiple of 11.
 - (x) 12 if the number is divisible by both 3 and 4.
 - (xi) 15 if the number is divisible by both 3 and 5.
 - (xii) 25 if the number formed by last two digits is divisible by 25.
- Product of factors of a given number is always equal to the number.
- $\text{HCF} = \text{Product of common factors of two or more numbers.}$
- $\text{LCM} = \text{Product of common and non-common factors.}$
- $\text{Product of two non-zero number} = \text{LCM} \times \text{HCF}$